

New bounds for kernel sums via fast spherical embeddings

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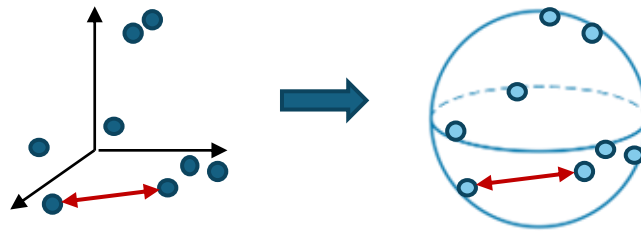
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Overview

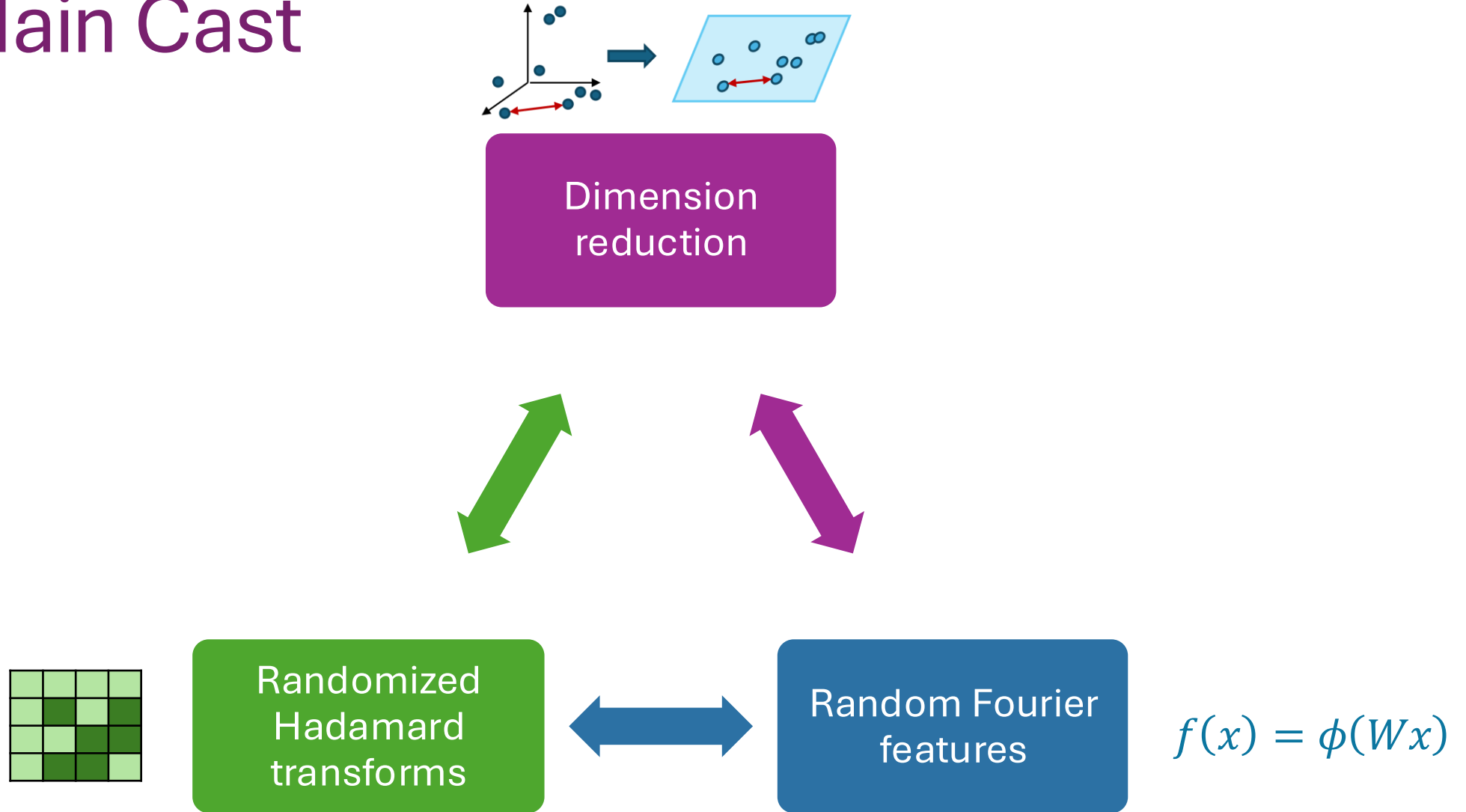
- Main theorem 1: Faster gaussian KDE approximation

$$y \mapsto \frac{1}{|X|} \sum_{x \in X} e^{-\|y-x\|_2^2} \pm \varepsilon$$

- Main theorem 2: Fast spherical embedding
 - Map high dimensional points to sphere in a distance-preserving way



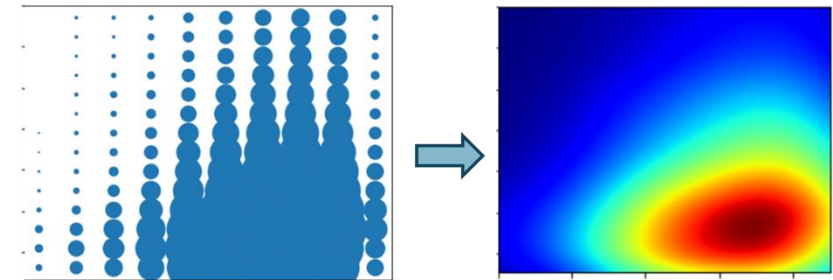
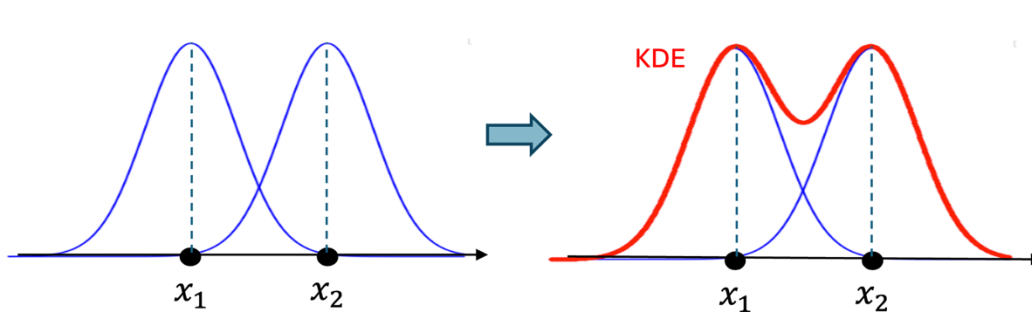
The Main Cast



Problem:

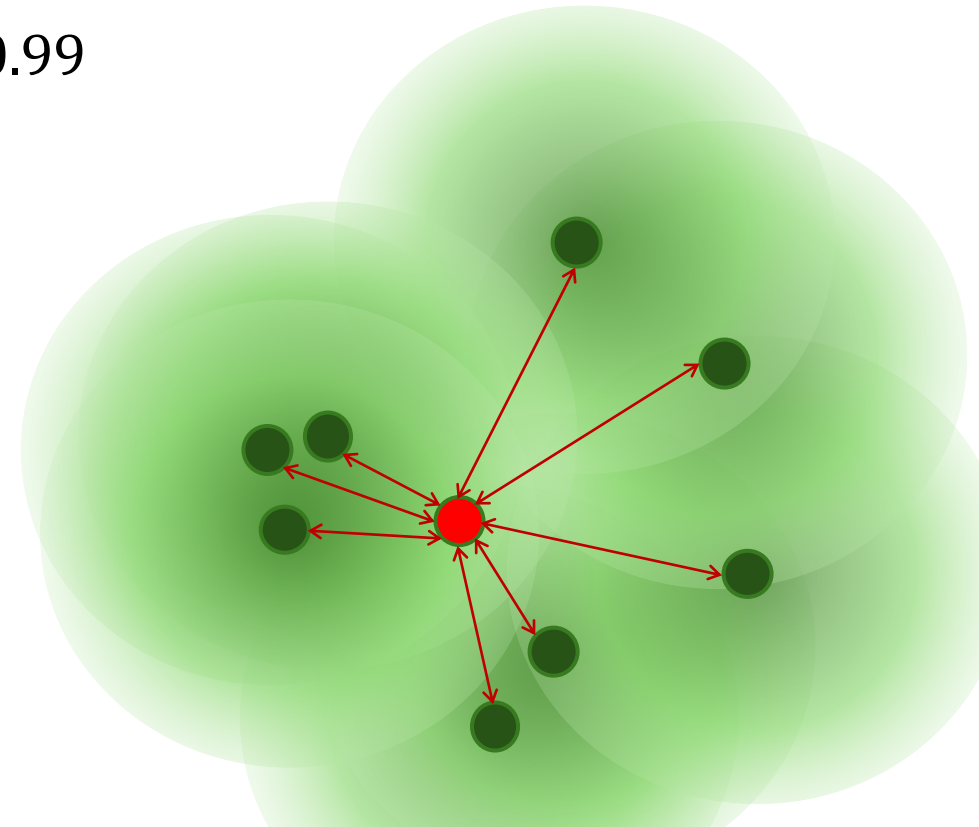
Gaussian Kernel Density Estimation

- Input: Finite (arbitrary) set of n points $X \subset \mathbb{R}^d$
- Task: Given a query $y \in \mathbb{R}^d$, return $KDE_X(y) := \frac{1}{|X|} \sum_{x \in X} e^{-\|y-x\|_2^2}$
- Goal: Compute this up to error $\pm \varepsilon$ as fast as possible
 - Exact computation time: $O(nd)$



Computational Problem: KDE Data Structures

- A KDE data structure has two operations:
 - **Build** (happens once in the beginning): Get $X \subset \mathbb{R}^d$, build data structure
 - **Query** (happens many times): Get $y \in \mathbb{R}^d$, return $KDE_X(y) \pm \varepsilon$
- Requirement: Each query succeeds with probability 0.99
- **Goal: Minimize query time**
- Assumptions:
 - **High dimension**: $\exp(d)$ not allowed
 - **Bounded range**: points have diameter Δ



Our Result

What query time can we get?

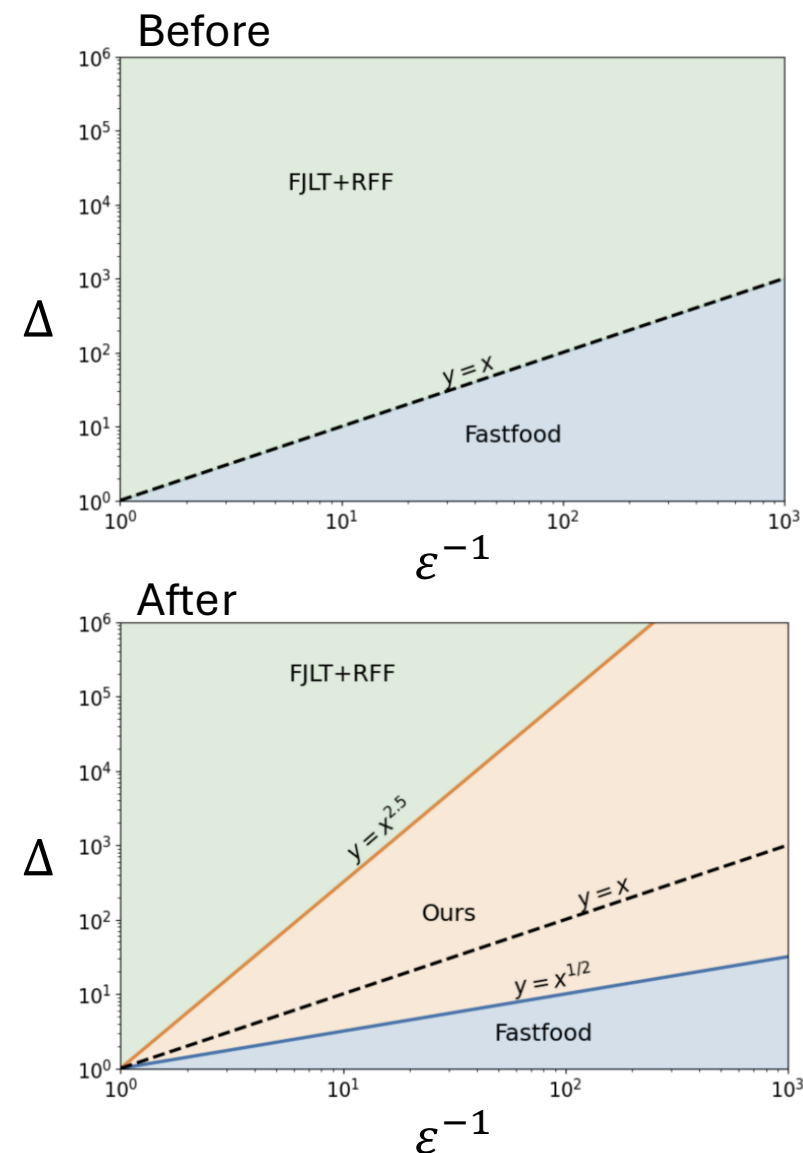
- Prior work:

- $\left(\log\left(\frac{1}{\varepsilon}\right)\right)^{O(d)}$ [Fast Gauss Transform]
- $O\left(\frac{d}{\varepsilon^2}\right)$ [Random Fourier Features]
- $\tilde{O}\left(d + \frac{1}{\varepsilon^4}\right)$ [Fast dimension reduction + RFF]
- $\tilde{O}\left(d + \frac{\Delta^2}{\varepsilon^2}\right)$ ["Fastfood"]

- Ours:

- $\tilde{O}\left(d + \varepsilon\Delta^2 + \frac{1}{\varepsilon^3}\right)$

Hypothesis:
 $\tilde{O}\left(d + \frac{1}{\varepsilon^2}\right) ???$



Blueprint: Linear Features for KDE

- KDE data structure blueprint:

- Find a feature map $f: \mathbb{R}^d \rightarrow \mathbb{R}^m$ such that $f(x)^T f(y) \approx e^{-\|y-x\|_2^2}$

- Build: Compute $F(X) = \frac{1}{|X|} \sum_{x \in X} f(x)$

- Query: Return $F(X)^T f(y) = \frac{1}{|X|} \sum_{x \in X} f(x)^T f(y) \approx \frac{1}{|X|} \sum_{x \in X} e^{-\|y-x\|_2^2} = KDE_X(y)$

- In low dimensions: Fast Gauss Transform [Greengard & Strain '91]:

- Get f via truncated Hermite (or Taylor) expansion (but has $\exp(d)$)

- In high dimensions?

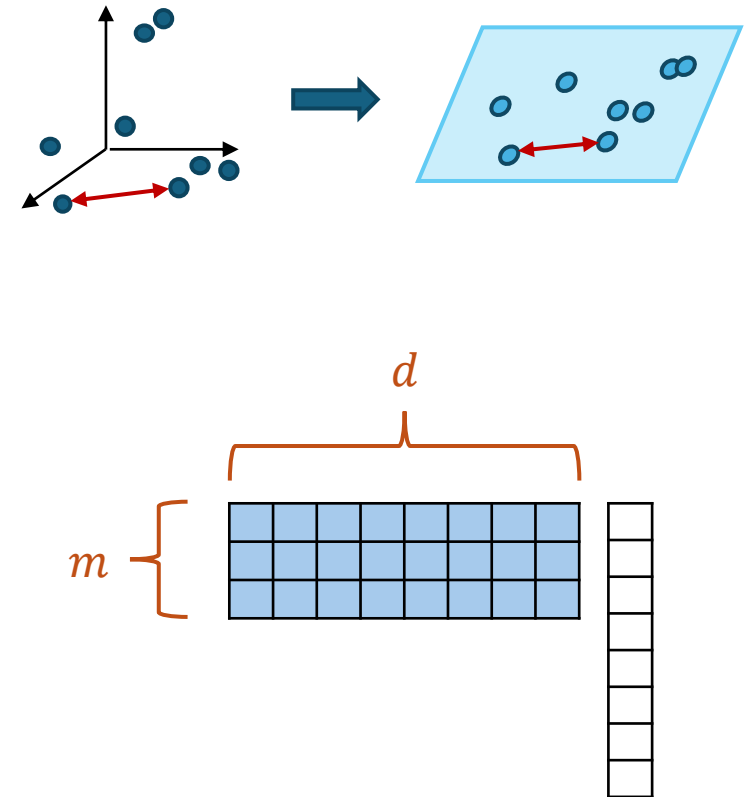
Random Fourier Features for KDE

- Goal: $f(x)^T f(y) \approx e^{-\|y-x\|_2^2}$
- Let $\phi: \mathbb{R}^m \rightarrow \mathbb{R}^{2m}$ be a mapping of vector to sin/cos pairs of its entries:
$$v \mapsto \bigoplus_{j=1}^m (\cos(v_j), \sin(v_j))$$
- **RFF** [Rahimi-Recht'07]: $f(x) = \phi(Wx)$ with iid gaussian $W \in \mathbb{R}^{m \times d}$
 - Need $m \approx 1/\varepsilon^2$, time: $O\left(\frac{d}{\varepsilon^2}\right)$
- How can we speed this up?

Dimension Reduction

[Johnson-Lindenstrauss'84]:

- We can reduce the dimension from d to $m \approx \frac{1}{\varepsilon^2}$ and preserve Euclidean distances up to $(1 \pm \varepsilon)$
- How? Multiply points by $W \in \mathbb{R}^{m \times d}$ with **i.i.d. gaussian entries**
- Time per point: $O\left(\frac{d}{\varepsilon^2}\right)$ -- **same as RFF**
- How can we speed this up?

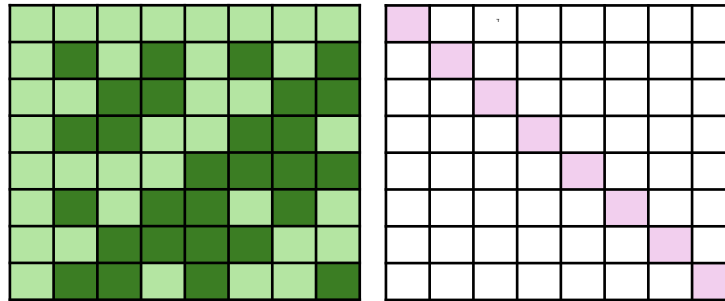


Randomized Hadamard Transform (RHT)

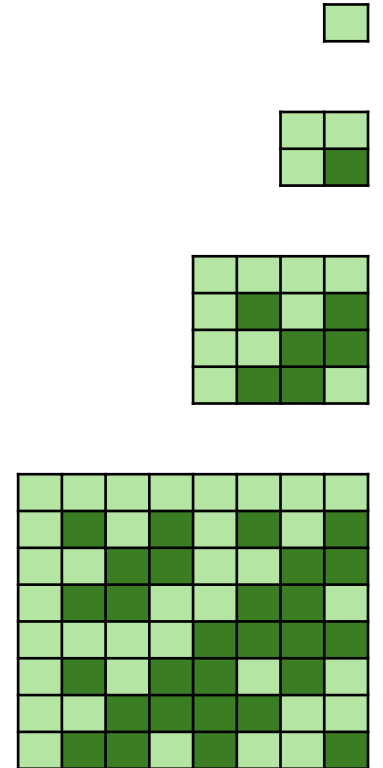
- The **Hadamard** matrix of order $2^\ell \times 2^\ell$ is defined by induction:

$$H_{2^0} = [1], \quad H_{2^\ell} = \frac{1}{\sqrt{2}} \begin{bmatrix} H_{2^{\ell-1}} & H_{2^{\ell-1}} \\ H_{2^{\ell-1}} & -H_{2^{\ell-1}} \end{bmatrix}$$

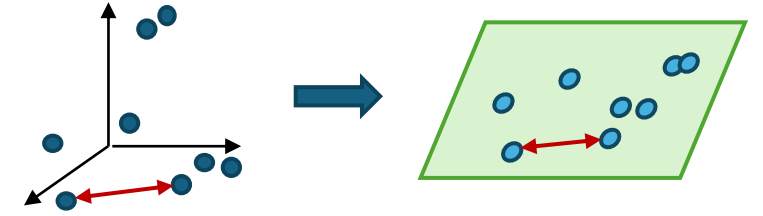
- An **RHT** is H times a random *diagonal* gaussian matrix D :



- Why?
 - HD can be multiplied by a vector in time $O(d \log d)$
 - HD behaves a lot like a **full gaussian matrix**



Fast Dimension Reduction: FJLT

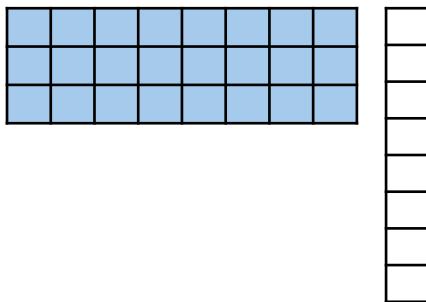


[Ailon-Chazelle'06]:

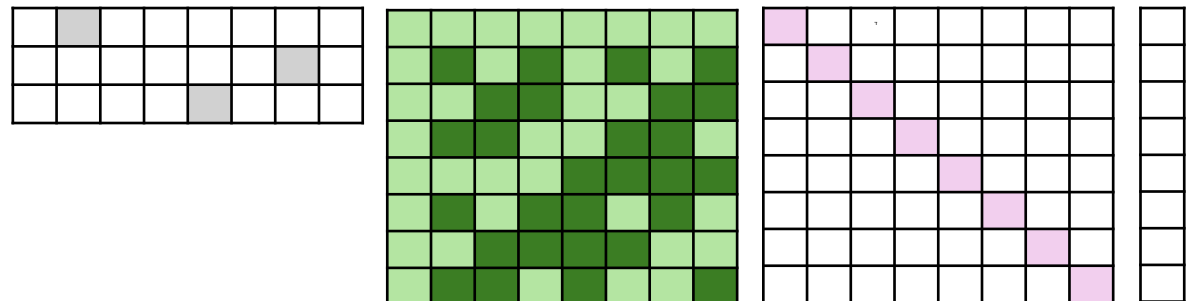
- RHT gives JL in time $\tilde{O}\left(d + \frac{1}{\varepsilon^2}\right)$
- How? Replace W with SHD
 - $S \in \mathbb{R}^{m \times d}$ is a row-sampling matrix

Same dimension reduction
guarantee as dense-
gaussian JL in faster time

JL

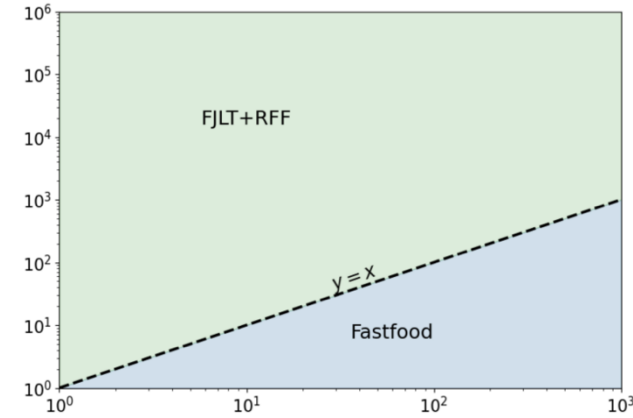


FJLT



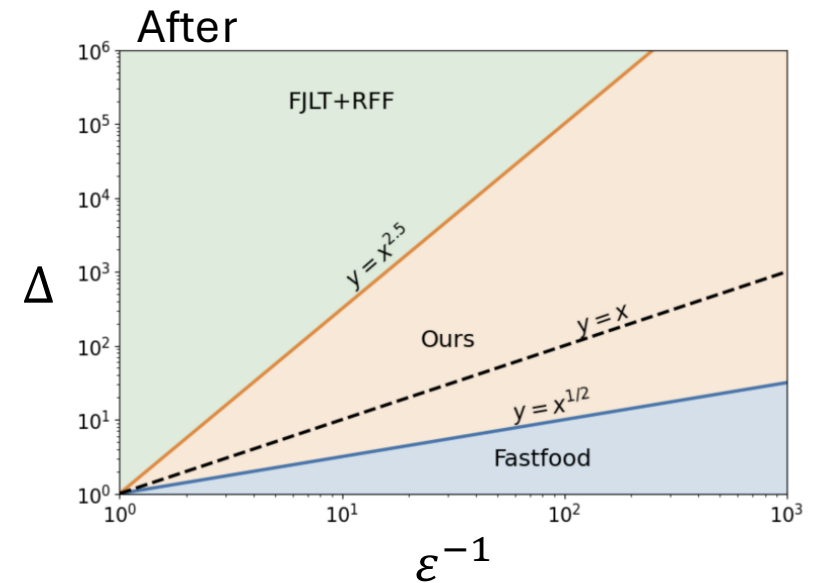
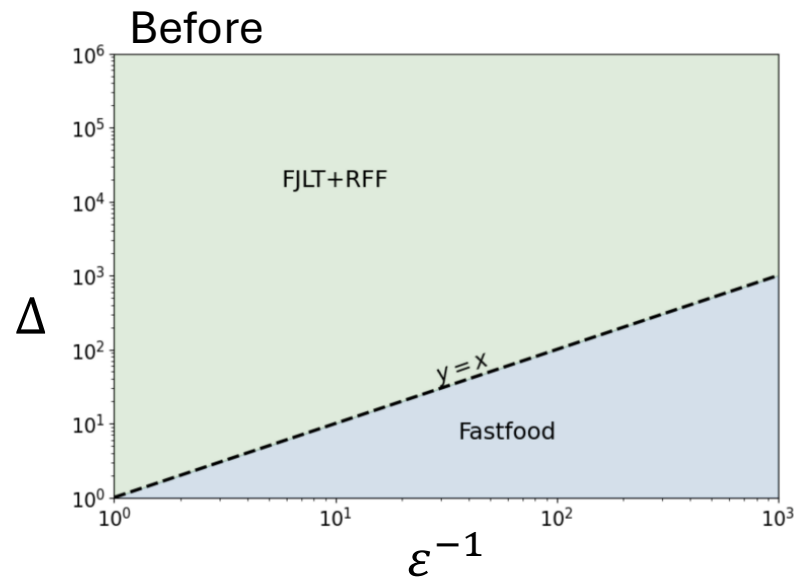
Faster RFF with RHT

- Goal: $f(x)^T f(y) \approx e^{-\|y-x\|_2^2}$
- RFF: $f(x) = \phi(Wx)$
- [Backurs et al. '24]: Use FJLT as preprocessing: $f(x) = \phi(W \cdot SHDx)$
 - Still needs iid gaussian W for RFF, but now it's $\approx \frac{1}{\varepsilon^2} \times \frac{1}{\varepsilon^2}$
 - Time: $\tilde{O}\left(d + \frac{1}{\varepsilon^4}\right)$
- [Le-Sarlos-Smola'13]: “Fastfood”: $f(x) = \phi(SHD_2 HD_1 x)$
 - Uses two iterated RHTs
 - There is theoretical and empirical literature showing iterated RHTs help
 - Cost: Incur dependence on the diameter, hence time: $\tilde{O}\left(d + \frac{\Delta^2}{\varepsilon^2}\right)$

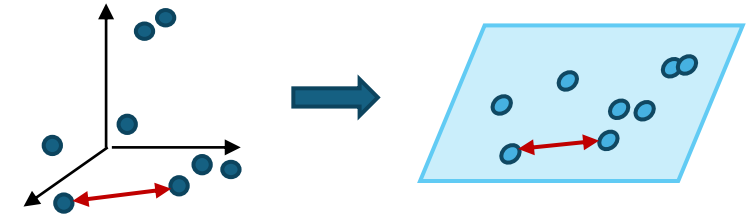


Our Plan of Action

1. Find a way to reduce the diameter Δ
2. Make it fast
3. Apply Fastfood

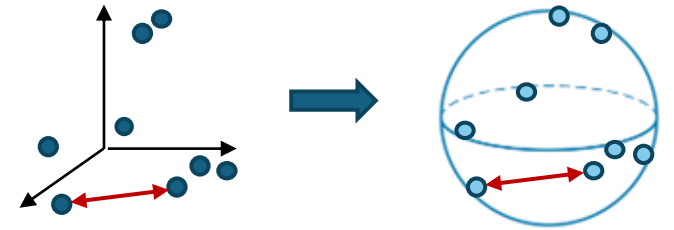


Dimension Reduction for KDE



- JL (or FJLT) is a map $g: \mathbb{R}^d \rightarrow \mathbb{R}^m$ preserves Euclidean distances up to $(1 \pm \varepsilon)$ w.h.p.:
 - **Non-Expansion:** $\|g(x) - g(y)\|_2^2 \leq (1 + \varepsilon)\|x - y\|_2^2$
 - **Non-Contraction:** $\|g(x) - g(y)\|_2^2 \geq (1 - \varepsilon)\|x - y\|_2^2$
- *But do we need to preserve all distances for KDE?*
 - Suppose $\|x - y\|_2^2 \geq \ln\left(\frac{1}{\varepsilon}\right)$
 - Then $e^{-\|x - y\|_2^2} \leq \varepsilon$, hence x contributes negligibly to $KDE_X(y)$
 - We just need it to stay negligible: $e^{-\|g(x) - g(y)\|_2^2} \leq \varepsilon$
 - So, we need **Non-Collapse:** $\|g(x) - g(y)\|_2^2 \geq \ln\left(\frac{1}{\varepsilon}\right)$
 - This is **weaker** than non-contraction, and should allow us to **limit diameter**

This Exists! [Bartal-Recht-Schulman'11]

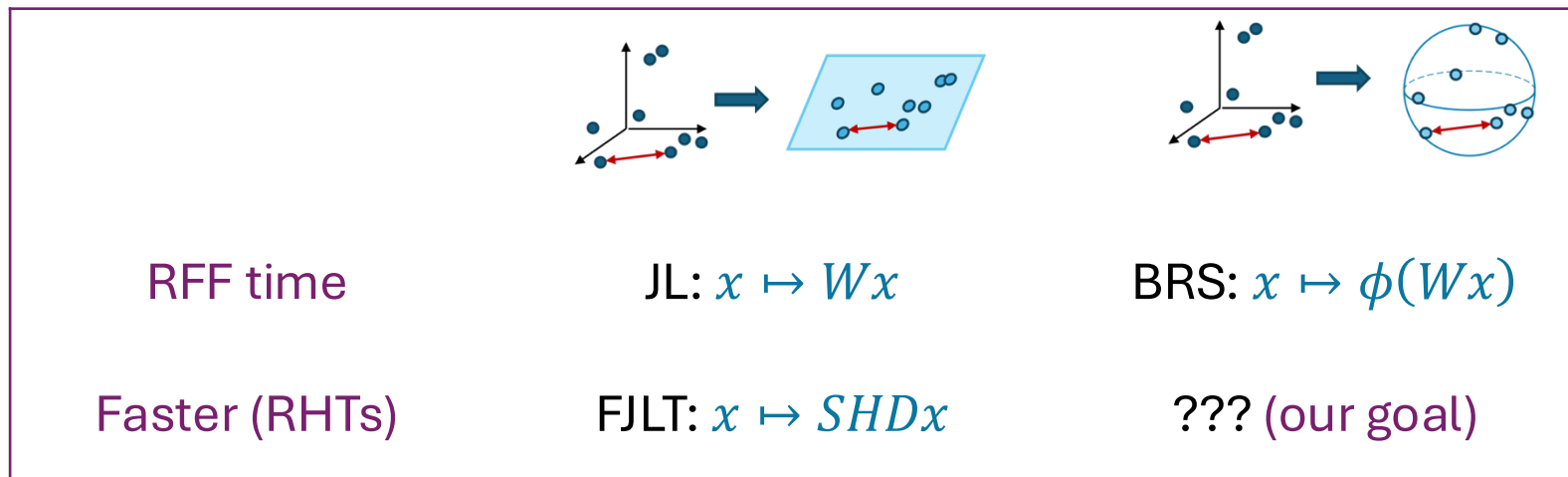


- Theorem - “Spherical dimension reduction”: We can embed the points into the **unit sphere** in dimension $m \approx \frac{1}{\varepsilon^2}$ such that for every x, y , w.h.p:
 - **Non-Expansion**: $\|g(x) - g(y)\|_2^2 \leq (1 + \varepsilon)\|x - y\|_2^2$
 - If $\|x - y\|_2^2 \leq \varepsilon$, **Non-Contraction**: $\|g(x) - g(y)\|_2^2 \geq (1 - \varepsilon)\|x - y\|_2^2$
 - If $\|x - y\|_2^2 > \varepsilon$, **Non-Collapse**: $\|g(x) - g(y)\|_2^2 \geq \Omega(\varepsilon)$
- Are we finished?
 - Scale the critical squared-distance threshold $\varepsilon \mapsto \ln(1/\varepsilon)$
 - This scales the squared-diameter to $\hat{\Delta}^2 = \frac{\ln(1/\varepsilon)}{\varepsilon}$
 - Apply Fastfood and get time $\tilde{O}\left(d + \frac{\hat{\Delta}^2}{\varepsilon^2}\right) = \tilde{O}\left(d + \frac{1}{\varepsilon^3}\right)$

But how much
time does it
take?

Faster Spherical Dimension Reduction?

- **Problem:** Spherical JL takes time $\left(\frac{d}{\varepsilon^2}\right)$ -- as slow as RFF
- In fact, the algorithm is identical to RFF: $g(x) = \phi(Wx)$
 - Different analysis of the same randomized map
- Can we make it faster with RHTs?

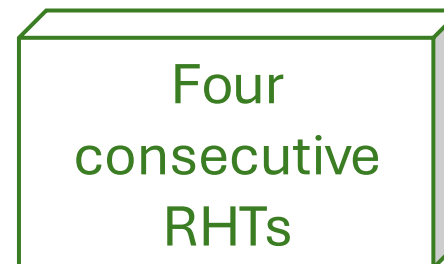


Our Theorem: Faster Spherical Embedding

But with fine-print

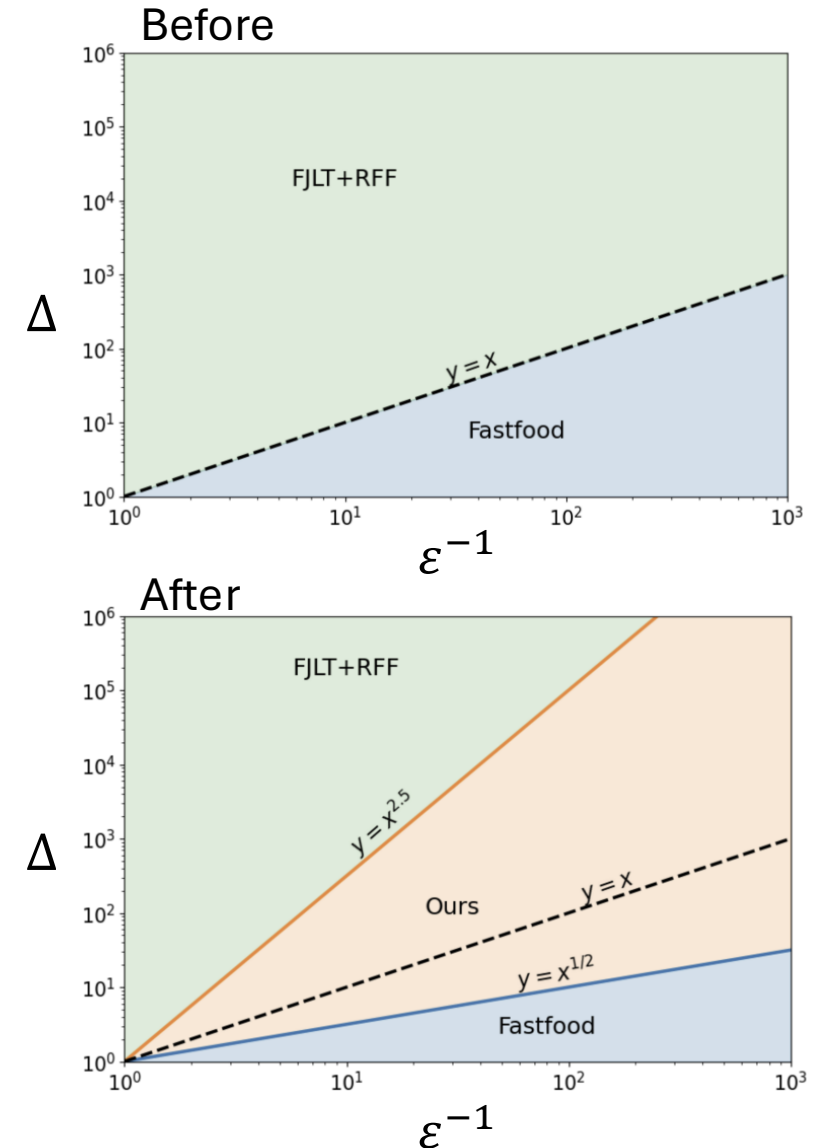
- We analyze $x \mapsto \phi(HD_2HD_1x)$ as fast spherical embedding
- Theorem: Let $\Lambda > 0$. We can embed the points into the unit sphere in dimension $m \approx \frac{1}{\varepsilon^2}$ and in time $\tilde{O}\left(d + \frac{1}{\varepsilon^2} + \Lambda^2\right)$ such that for every x, y , w.h.p:
 - Non-Expansion: $\|g(x) - g(y)\|_2^2 \leq (1 + \varepsilon)\|x - y\|_2^2$
 - If $\|x - y\|_2^2 \leq \varepsilon$, Non-Contraction: $\|g(x) - g(y)\|_2^2 \geq (1 - \varepsilon)\|x - y\|_2^2$
 - If $\Lambda^2 \geq \|x - y\|_2^2 > \varepsilon$, Non-Collapse: $\|g(x) - g(y)\|_2^2 \geq \varepsilon$

- Our final KDE features: $f(x) = \underbrace{\phi(SHD_4HD_3)}_{\text{Fastfood}} \cdot \underbrace{\phi(HD_2HD_1x)}_{\text{Our new spherical embedding}}$



Our Theorem: Faster KDE

- Scale the critical distance² from $\ln(1/\varepsilon)$ to ε
- Input diameter² to fast spherical embedding: $\approx \varepsilon \Delta^2$
- Fast spherical embedding takes time: $\tilde{O}\left(d + \frac{1}{\varepsilon^2} + \varepsilon \Delta^2\right)$
- Output diameter² from fast spherical embedding: $\approx 1/\varepsilon$
- Fastfood takes time: $\tilde{O}\left(d + \frac{1}{\varepsilon^3}\right)$
- **Total time:** $\tilde{O}\left(d + \frac{1}{\varepsilon^3} + \varepsilon \Delta^2\right)$



Gist of Analysis: Concentration for RHTs

- Our spherical embedding map is $\Phi(x) = \phi(HD_2HD_1x)$
- By trig identities: $\|\Phi(x) - \Phi(y)\|_2^2 = \frac{1}{m} \sum_{j=1}^m (1 - \cos G_j)$ where $G_j \sim N(0, \|x - y\|_2^2)$
- But the G_j s are **dependent**: they are polynomials in the entries of D_1 and D_2

$$G = H D_2 H D_1 (x - y)$$

- Main tool: **Wiener chaos decomposition** into hypercontractive terms
- We need a **4th chaos analysis** because $\frac{1}{2}G^2 - \frac{1}{24}G^4 \leq 1 - \cos G \leq \frac{1}{2}G^2$

Conclusion

- Main results:

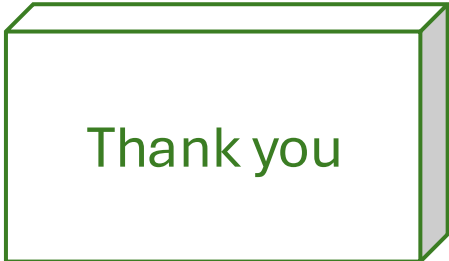
- New bound for gaussian KDE approximation up to $\pm \varepsilon$
- New fast spherical embedding theorem (FJT-analog of Bartal-Recht-Schulman)

- Extensions:

- Extends to other kernels, e.g., inverse multi-quadratic: $k(x, y) = \left(\frac{1}{1 + \|x - y\|_2^2} \right)^\beta$
- Extends to other settings, e.g., differential privacy
 - Five RHTs! $f_{DP}(x) = HD_5 \cdot \phi(SHD_4 HD_3 \cdot \phi(HD_2 HD_1 x))$

- Open questions:

- Optimal bound for gaussian KDE? $\tilde{O}\left(d + \frac{1}{\varepsilon^2}\right) ???$
- More applications of fast spherical embedding?



Thank you